BHARATIYA ANTARIKSH HACKATHON 2026

APEX-ORACLE

Physics-aware, explainable AI for detecting exoplanets in noisy TESS light curves

Problem Statement AI-enabled Detection of Exoplanets from Noisy Astronomical Light Curves

Team [YOUR TEAM NAME] **Team Leader** [YOUR NAME] (solo entry)

THE TEAM

A solo entry, full-stack ownership

YN

[YOUR NAME]

Solo participant - ML engineering + astro-informatics

One person, every hat in the pipeline:

- **Data & detrending**
- **Transformer modeling**
- **Bayesian physics**
- **False-positive vetting**
- **Explainability (XAI)**
- **Live demo & report**

THE OPPORTUNITY

Why APEX-ORACLE is different

1

Different from existing ideas

Most entries are black-box CNNs that only flag a brightness dip. APEX-ORACLE also classifies the signal, returns physical parameters with confidence, and shows visual reasons for every call.

2

How it solves the problem

GP detrending preserves shallow transits, a multimodal model plus centroid vetting rejects blended false positives, and a differentiable transit model recovers parameters - even single transits.

3

Unique selling point

Detection + classification + Bayesian parameters + explainability in one validated pipeline, proven by injection-recovery tests. Accurate and fully transparent - never a black box.

CAPABILITIES

What the solution does

Transit-safe detrending

GP + Savitzky-Golay removes noise, keeps 100 ppm dips

4-class signal typing

Transit, eclipsing binary, blend, or starspot

U-vs-V disentanglement

Attention learns transit vs binary geometry

Differentiable parameters

Period, depth, duration with calibrated bounds

False-positive vetting

Centroid / difference imaging kills blends

Monotransit recovery

Finds single-transit, long-period planets

Explainable visuals

Phase-folded curves with attention overlays

Honest validation

Injection-recovery completeness & SNR

PROCESS FLOW

Nine-stage detection pipeline

1. **Ingestion**
   lightkurve, astroquery
2. **Detrending**
   wotan, celerite2
3. **Candidate + fold**
   transitleastsquares
4. **Classification**
   Transformer + XGBoost
5. **Parameter estimation**
   diff. batman + sbi
6. **Vetting**
   centroid, triceratops
7. **Explainability**
   attention, matplotlib
8. **Validation**
   injection-recovery
9. **Live demo**
   Streamlit dashboard

MOCK-UP (OPTIONAL)

Live inspector dashboard

APEX-ORACLE - Exoplanet Inspector

TIC 307210830

Analyze

Phase-folded light curve + attention

TRANSIT confidence 96%

Period: **3.41 d +/- 0.02**

Depth: **910 ppm +/- 40**

Duration: **2.7 h +/- 0.1**

Vetting: **on-target, FPP 0.4%**

ARCHITECTURE

APEX-ORACLE hybrid pipeline

Data & labels

TESS / MAST,
Gaia DR3, TOI

Detrend + BLS/TLS

transit-safe GP,
candidate search

A

Transformer

multimodal
classifier

B

Diff. batman + SBI

Bayesian
parameters

C

Vetting + output

centroid, FPP,
XAI overlays

Reproducibility & infra: uv - Hydra / OmegaConf - Weights & Biases - PyTorch Lightning - injection-recovery harness

A = fast front-end . B = detection spine . C = parameter & confidence head

TECHNOLOGY STACK

Technologies used

Data & astronomy

lightkurve - astropy - astroquery

transitleastsquares - wotan

Detrending & GP

wotan - celerite2 - gpytorch

Deep learning

PyTorch - Lightning - timm

Transformers - XGBoost

Physics & Bayesian

batman - sbi - dynesty - emcee

Vetting & XAI

triceratops - attention maps

matplotlib - plotly

Infra & delivery

uv - Hydra - Weights & Biases

Streamlit

Language: Python 3.11 . Compute: GPU for training, CPU-capable inference

ESTIMATED COST (OPTIONAL)

Near-zero, fully reproducible

Data	TESS archive (MAST)	**Free**
Software	100% open-source stack	**Free**
Compute	Kaggle / Colab GPU, or short cloud-GPU rental	**Rs 0 - 8,000**

TOTAL

~ Rs 0

to a few thousand

Low-risk and deployable: open data, open tools, runs on free-tier GPUs - the cost is engineering effort, not capital.

APEX-ORACLE

Detect. Explain. Trust.

Finding new worlds in the noise - and proving every discovery.

Bharatiya Antariksh Hackathon 2026 . [YOUR NAME] . [YOUR TEAM NAME]